

Proving the Fundamental Theorem of Algebra

Let $\{a_k\}_{k=0}^n \subset \mathbb{C}$, where $a_n \neq 0$. With this collection of complex numbers, construct the n^{th} degree polynomial function $p : \mathbb{C} \rightarrow \mathbb{C}$ defined as such:

$$p(z) = \sum_{k=0}^n a_k z^k \quad \forall z \in \mathbb{C}$$

Now suppose there exists a set $\mathcal{Z} = \{z_k\}_{k=1}^N$ such that $p(z_k) = 0$ for each k . It goes without saying that if $p(z) \neq 0, \forall z \in \mathbb{C}$, then $N = 0$, and $\mathcal{Z} = \emptyset$.

Furthermore, by the Factor Theorem, for some polynomial function $q : \mathbb{C} \rightarrow \mathbb{C} \setminus \{0\}$, $\exists \{m_k\}_{k=1}^N \subset \mathbb{N}$ such that p could be expressed as such:

$$p(z) = q(z) \prod_{k=1}^N (z - z_k)^{m_k} \quad \text{with the understanding} \quad N = 0 \implies p(z) = q(z)$$

By definition, $\deg[p] = n$, which implies upon inspection, $n = \deg[p] = \deg[q] + \sum_{k=1}^N m_k$. As $\deg[q] \geq 0$ and $m_k \geq 1$, it follows that $n \geq N$.

For any function $f \in C^1(\mathbb{C}, \mathbb{C} \setminus \{0\})$, the logarithmic derivative of f is defined to be f'/f . Provided that g is of the same class as f , the logarithmic derivative of fg is given by $(fg)'/(fg) = (f'g + fg')/(fg) = f'/f + g'/g$. It follows (by induction) that the logarithmic derivative of p is given as follows:

$$\frac{p'(z)}{p(z)} = \frac{q'(z)}{q(z)} + \sum_{k=1}^N \frac{\frac{d}{dz}(z - z_k)^{m_k}}{(z - z_k)^{m_k}} = \frac{q'(z)}{q(z)} + \sum_{k=1}^N \frac{m_k}{z - z_k} \quad \forall z \in \mathbb{C} \setminus \mathcal{Z}$$

As $N \leq n < \infty$, $\max_{z \in \mathcal{Z}} |z| \in \mathbb{R}^+$ is well defined. Now, let C_R be a positively oriented contour defined by $C_R = \{z : z \in \mathbb{C}, |z| = R\}$, where $R > \max_{z \in \mathcal{Z}} |z|$. Thus, by construction, C_R encloses $\mathcal{Z}$. Additionally, because q'/q is entire, by Cauchy's Theorem and Residue Theorem, the following holds:

$$\oint_{C_R} \frac{p'(z)}{p(z)} dz = \oint_{C_R} \frac{q'(z)}{q(z)} dz + \sum_{k=1}^N m_k \oint_{C_R} \frac{1}{z - z_k} dz = 2\pi i \sum_{k=1}^N m_k$$

At this point, define the functional $\mathcal{N}_0[p] = \sum_{k=1}^N m_k$, which counts the number of zeros of p including multiplicities. From this point of view:

$$n = \deg[q] + \mathcal{N}_0[p] \quad \text{and} \quad \mathcal{N}_0[p] = \frac{1}{2\pi i} \oint_{C_R} \frac{p'(z)}{p(z)} dz$$

Now, by invoking the non-factorized expression of p , the total number of zeros of p counting multiplicities can likewise be computed as such:

$$\mathcal{N}_0[p] = \frac{1}{2\pi i} \oint_{C_R} \frac{\frac{d}{dz} \sum_{k=0}^n a_k z^k}{\sum_{k=0}^n a_k z^k} dz = \frac{1}{2\pi i} \oint_{C_R} \frac{\sum_{k=1}^n k a_k z^{k-1}}{\sum_{k=0}^n a_k z^k} dz$$

In order to compute the above integral, parameterize C_R by the map $\theta \in [0, 2\pi) \mapsto Re^{i\theta}$. In this respect, the dummy variable z can be expressed as $Re^{i\theta}$. Likewise, the differential element dz can thus be expressed as $iRe^{i\theta}d\theta$ in these coordinates:

$$\implies \mathcal{N}_0[p] = \frac{1}{2\pi i} \int_0^{2\pi} \frac{\sum_{k=1}^n k a_k (Re^{i\theta})^{k-1}}{\sum_{k=0}^n a_k (Re^{i\theta})^k} iRe^{i\theta} d\theta = \frac{1}{2\pi} \int_0^{2\pi} \frac{\sum_{k=1}^n k a_k R^{k-n} e^{ik\theta}}{\sum_{k=0}^n a_k R^{k-n} e^{ik\theta}} d\theta$$

Furthermore, the integral will be split into its constituent real and imaginary parts in order for real analytic techniques to be applied:

$$\implies \mathcal{N}_0[p] = \frac{1}{2\pi} \int_0^{2\pi} \operatorname{Re} \left(\frac{\sum_{k=1}^n k a_k R^{k-n} e^{ik\theta}}{\sum_{k=0}^n a_k R^{k-n} e^{ik\theta}} \right) d\theta + \frac{i}{2\pi} \int_0^{2\pi} \operatorname{Im} \left(\frac{\sum_{k=1}^n k a_k R^{k-n} e^{ik\theta}}{\sum_{k=0}^n a_k R^{k-n} e^{ik\theta}} \right) d\theta$$

By the Mean Value Theorem, $\exists \alpha_R, \beta_R \in [0, 2\pi)$ such that the above integral(s) can be expressed as follows:

$$\implies \mathcal{N}_0[p] = \operatorname{Re} \left(\frac{\sum_{k=1}^n k a_k R^{k-n} e^{ik\alpha_R}}{\sum_{k=0}^n a_k R^{k-n} e^{ik\alpha_R}} \right) + i \operatorname{Im} \left(\frac{\sum_{k=1}^n k a_k R^{k-n} e^{ik\beta_R}}{\sum_{k=0}^n a_k R^{k-n} e^{ik\beta_R}} \right)$$

In order to observe the asymptotic behavior of this expression as $R \rightarrow \infty$, Big O notation will be employed in powers of R^{-1} as such:

$$\implies \mathcal{N}_0[p] = \operatorname{Re} \left(\frac{na_n e^{in\alpha_R} + \mathcal{O}(R^{-1})}{a_n e^{in\alpha_R} + \mathcal{O}(R^{-1})} \right) + i \operatorname{Im} \left(\frac{na_n e^{in\beta_R} + \mathcal{O}(R^{-1})}{a_n e^{in\beta_R} + \mathcal{O}(R^{-1})} \right)$$

Because the above expression does not have explicit dependence on R (so long as R is large enough to encircle all the zeros of p), taking the limit as $R \rightarrow \infty$ will maintain overall equality:

$$\begin{aligned} \implies \mathcal{N}_0[p] &= \lim_{R \rightarrow \infty} \left(\operatorname{Re} \left(\frac{na_n e^{in\alpha_R} + \mathcal{O}(R^{-1})}{a_n e^{in\alpha_R} + \mathcal{O}(R^{-1})} \right) + i \operatorname{Im} \left(\frac{na_n e^{in\beta_R} + \mathcal{O}(R^{-1})}{a_n e^{in\beta_R} + \mathcal{O}(R^{-1})} \right) \right) \\ &= \operatorname{Re} \left(\lim_{R \rightarrow \infty} \frac{na_n e^{in\alpha_R} + \mathcal{O}(R^{-1})}{a_n e^{in\alpha_R} + \mathcal{O}(R^{-1})} \right) + i \operatorname{Im} \left(\lim_{R \rightarrow \infty} \frac{na_n e^{in\beta_R} + \mathcal{O}(R^{-1})}{a_n e^{in\beta_R} + \mathcal{O}(R^{-1})} \right) \\ &= \operatorname{Re} \left(\frac{na_n e^{in\alpha_\infty}}{a_n e^{in\alpha_\infty}} \right) + i \operatorname{Im} \left(\frac{na_n e^{in\beta_\infty}}{a_n e^{in\beta_\infty}} \right) = n \end{aligned}$$

By the string of equalities, it follows that $\mathcal{N}_0[p] = n$. Therefore, it directly follows that $\deg[q] = 0$, which further implies that q is constant. In this respect, by the distributive property ($\forall z \in \mathbb{C}$):

$$p(z) = q(z) \prod_{k=1}^N (z - z_k)^{m_k} = q(z) z^n + \sum_{k=0}^{n-1} a_k z^k \implies q(z) = a_n \quad \square$$

This concludes the proof of the Fundamental Theorem of Algebra. The theorem states that every n^{th} degree polynomial with complex valued coefficients has exactly n roots counting multiplicities. With the established context at hand, the theorem equivalently states:

$$\sum_{k=0}^n a_k z^k = a_n \prod_{k=1}^N (z - z_k)^{m_k} \quad \forall z \in \mathbb{C} \quad \text{where} \quad \sum_{k=1}^N m_k = n$$